



AshTech

Kids' Handbook to HTML & CSS

A picture guide for young web builders

Made for the students of Sir Syed School & College Campus-V,
and curious trainers everywhere.

What's inside

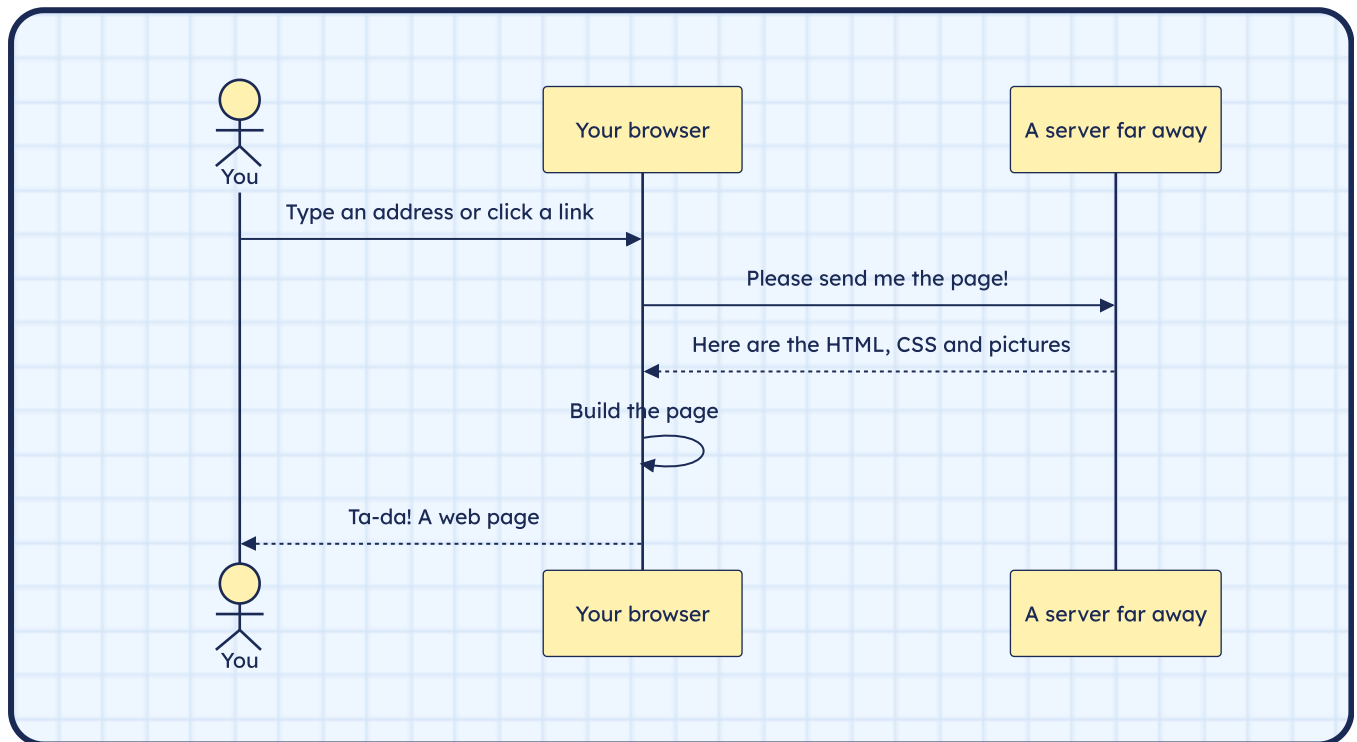
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What is a website?

A website is a set of files that live on a computer far away, called a **server**.

When you visit a website, your **browser** (Chrome, Edge, Safari...) asks the server for those files, then builds the page for you. It happens in less than a second!

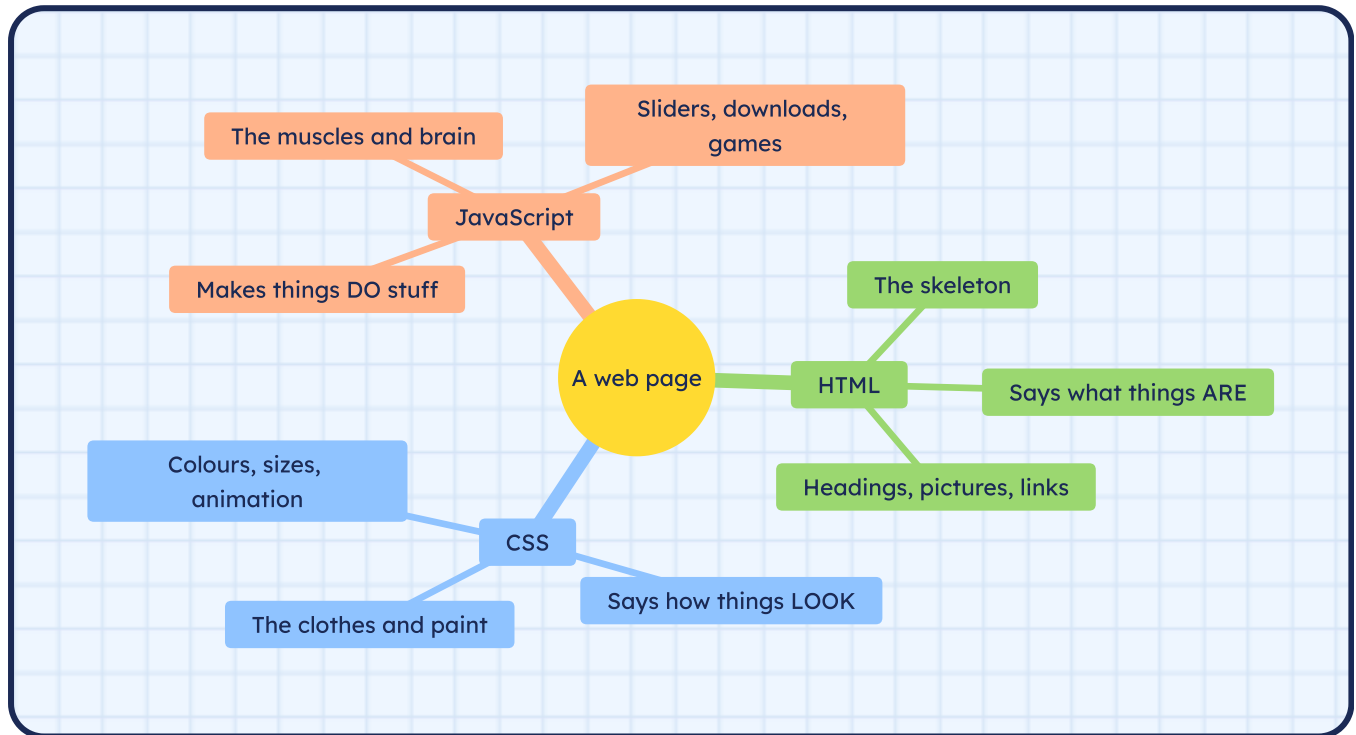


Remember: the browser is like a chef. The files are the recipe. The web page is the meal.



The three languages of the web

Web pages are written in three languages. Each one has its own job. AshTech is made almost entirely of the first two!

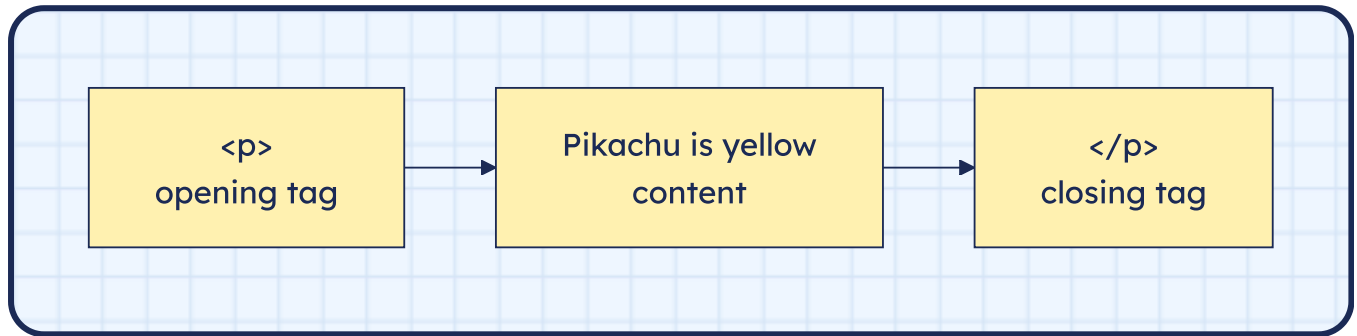


Remember: HTML = what it is. **CSS** = how it looks. **JavaScript** = what it does.



Your first tag

HTML is made of **tags**. A tag is a word inside angle brackets, like `<p>`. Most tags come in pairs: one to **open** and one to **close**. The closing tag has a slash `/`.



Tags can also have **attributes**: extra information inside the opening tag.

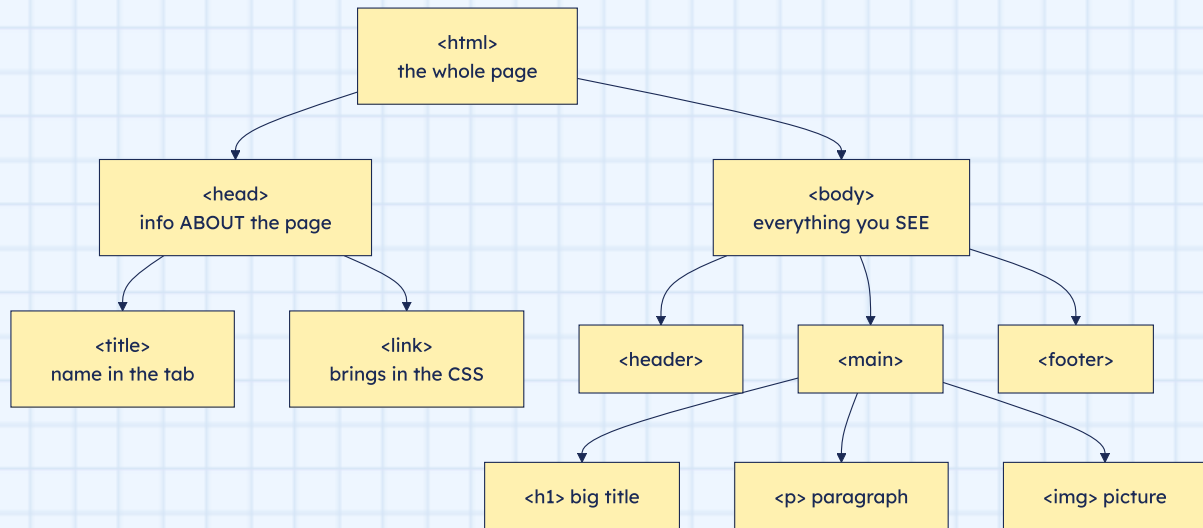
```
<a href="pokedex.html">Open the Pokédex</a>
```

Try it: in the Workshop's Code Lab, change the words between `<h2>` and `</h2>` to your own name.



The skeleton of every page

Every web page has the same bones. Tags go **inside** other tags, like boxes inside boxes. We call this **nesting**.



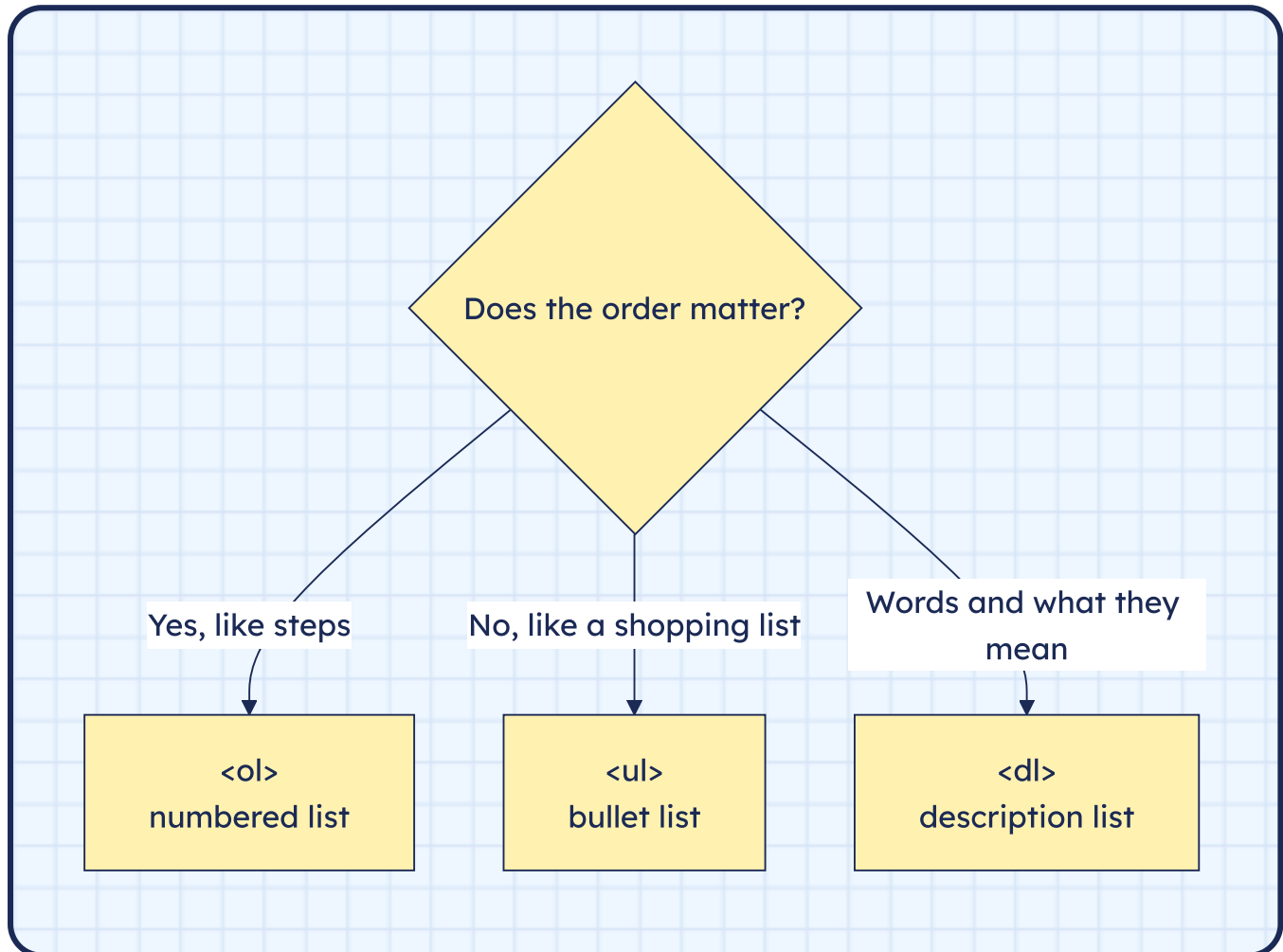
Remember: close tags in the opposite order you opened them. <p>

Hi</p> ✓ <p>Hi</p></p> ✗



Lists, links and pictures

Three tags you will use all the time. Lists keep things tidy, links take you places, and pictures make everything more fun.



```
<ul>
  <li>Bulbasaur</li>
  <li>Charmander</li>
</ul>

<a href="types.html">Learn the types</a>

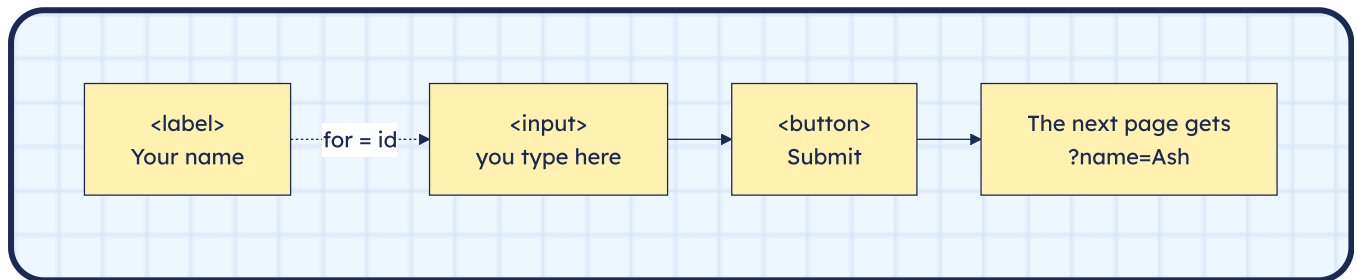

```

Remember: always give pictures alt words. People who can't see the picture hear the alt words read aloud instead.



Forms: asking questions

Forms let people type answers, tick boxes and press buttons. The Academy page is one big form! Every box needs a `<label>`, joined to it with matching `for` and `id`.



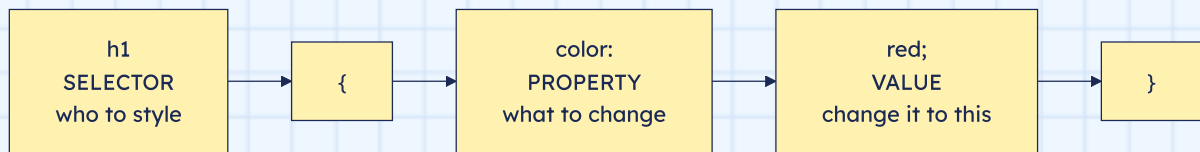
Some input types you can try: `text`, `number`, `date`, `color`, `range`, `checkbox` and `radio`.

Try it: fill in the Academy form, then look at the address bar after you submit. Can you find your name in it?

Meet CSS



CSS changes how things look. A CSS **rule** has a **selector** (who to style), then curly brackets holding **properties** (what to change) and **values** (what to change them to).



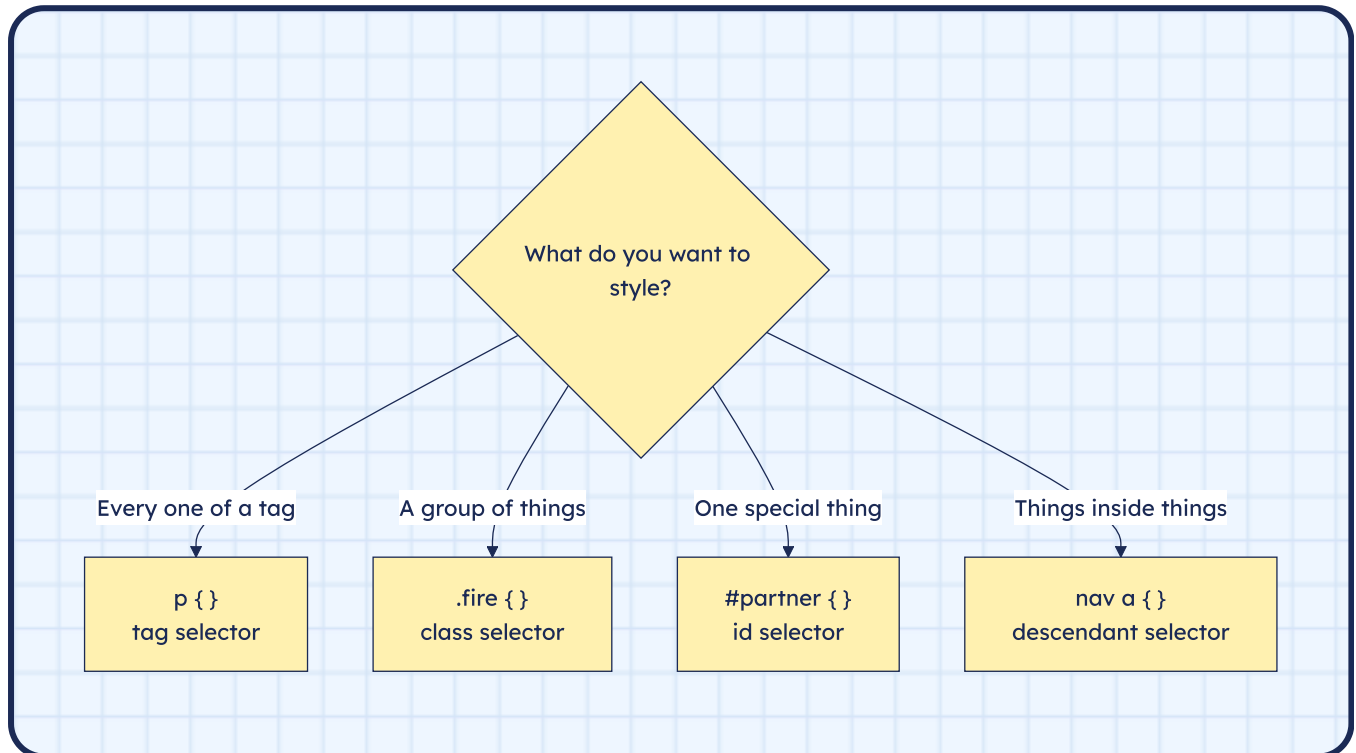
```
h1 {  
  color: red;  
  font-size: 40px;  
}
```

Remember: every line inside the curly brackets ends with a semicolon ; . Forgetting one is the number one CSS mistake!



Choosing what to style

Selectors are how CSS points at things. There are lots of them, but these four do most of the work.

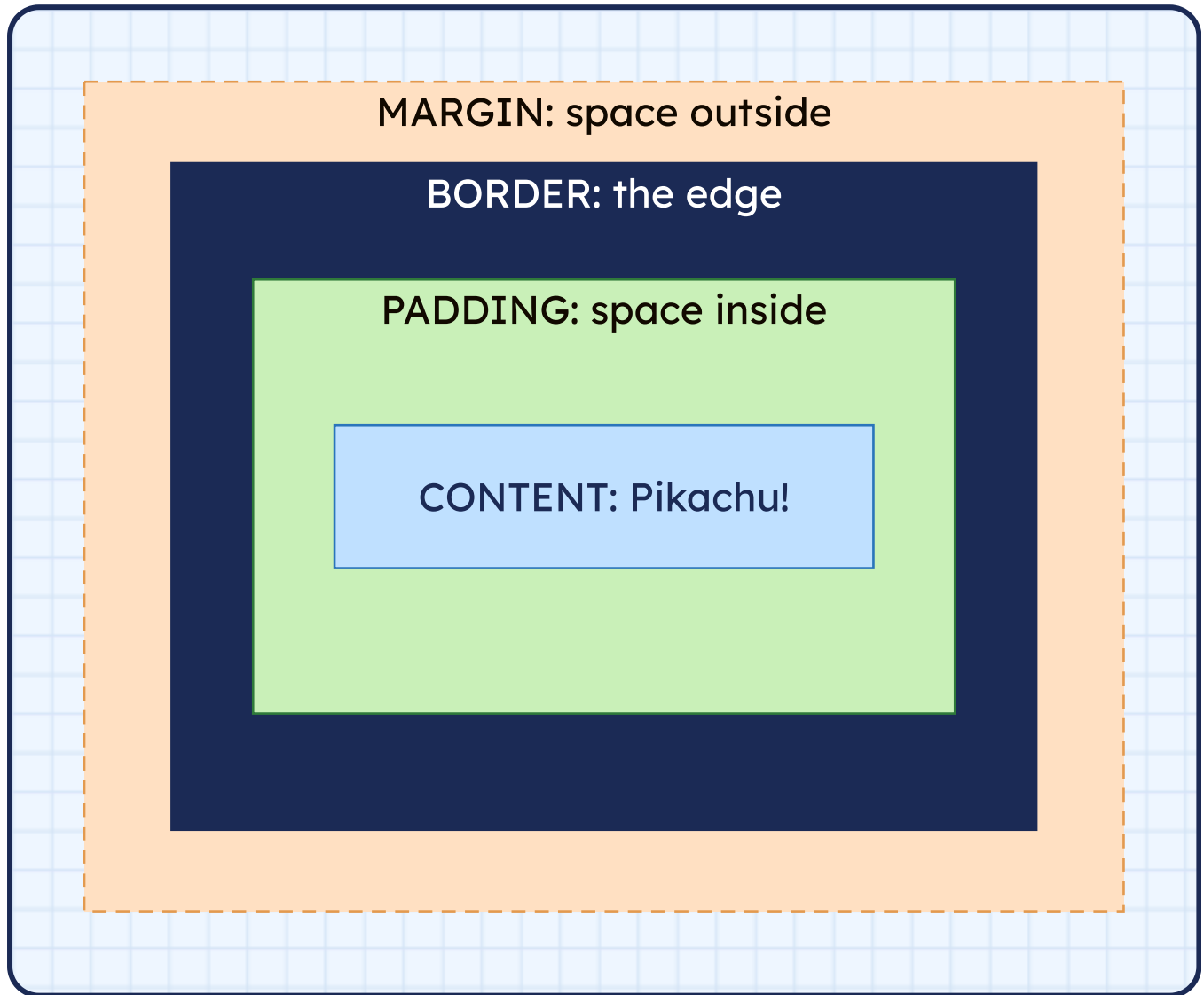


Try it: play Selector Detective in the Workshop and solve all eight missions.



Every element is a box

Everything on a web page is a box, even round things! Each box has four layers, like a Squirtle in its shell.

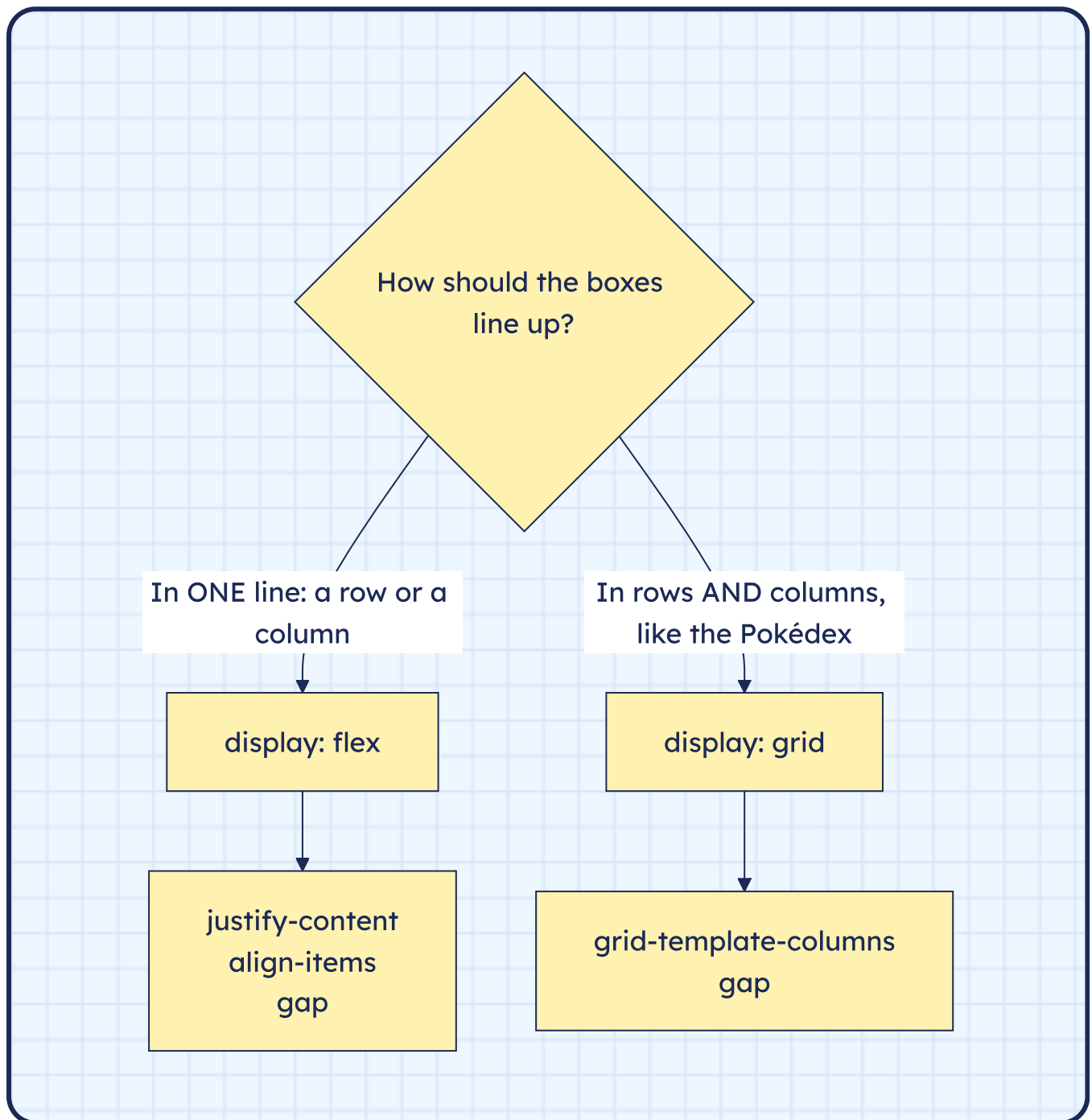


Try it: slide the numbers in Measure the box and watch the maths change.

Flexbox and grid: lining things up



CSS has two super tools for putting boxes where you want them. How do you choose?

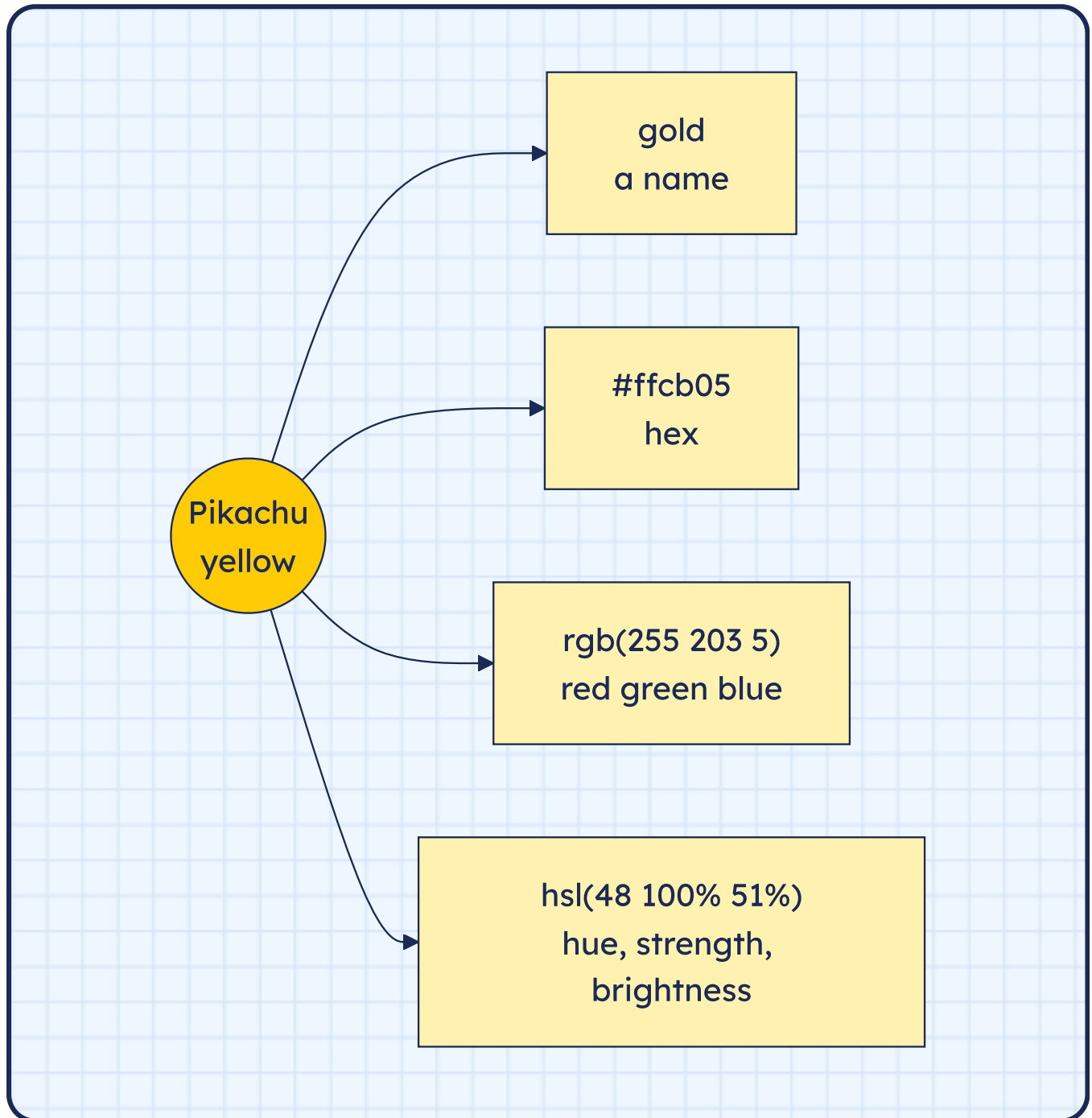


```
.team {  
  display: flex;  
  justify-content: center;  
  gap: 12px;  
}
```



Colours and fonts

You can write the same colour in different ways. They all mean exactly the same yellow!



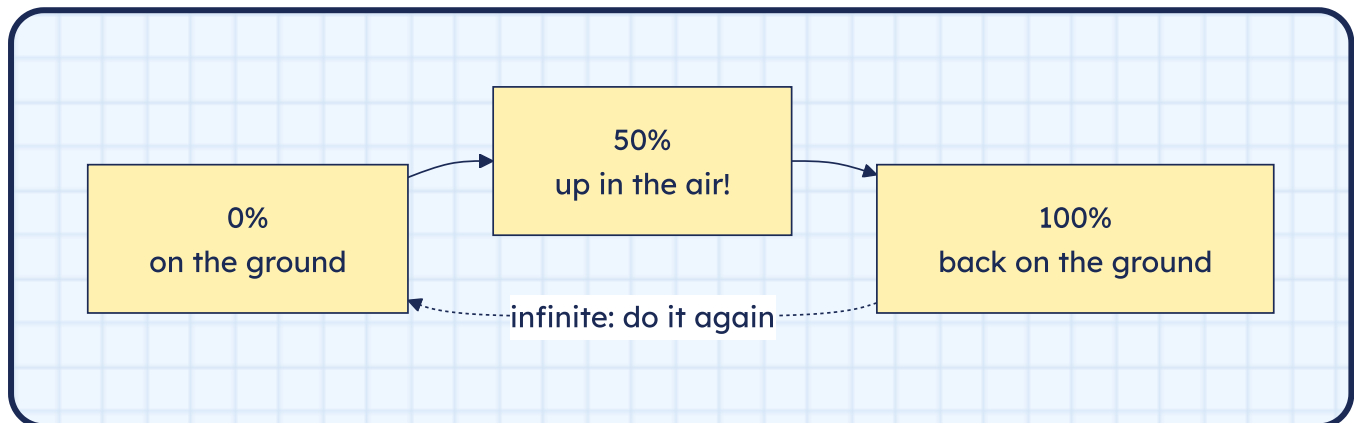
Fonts are just as easy: `font-family` picks the letters, `font-size` makes them bigger or smaller, and `font-weight: bold` makes them thick.

Try it: play “Match the type!” in the Colour Mixer. Can you score 95%?



Making things move

An animation is a list of steps called **keyframes**. You say what the Pokémon looks like at the start (0%), the middle (50%) and the end (100%). The browser fills in everything between!

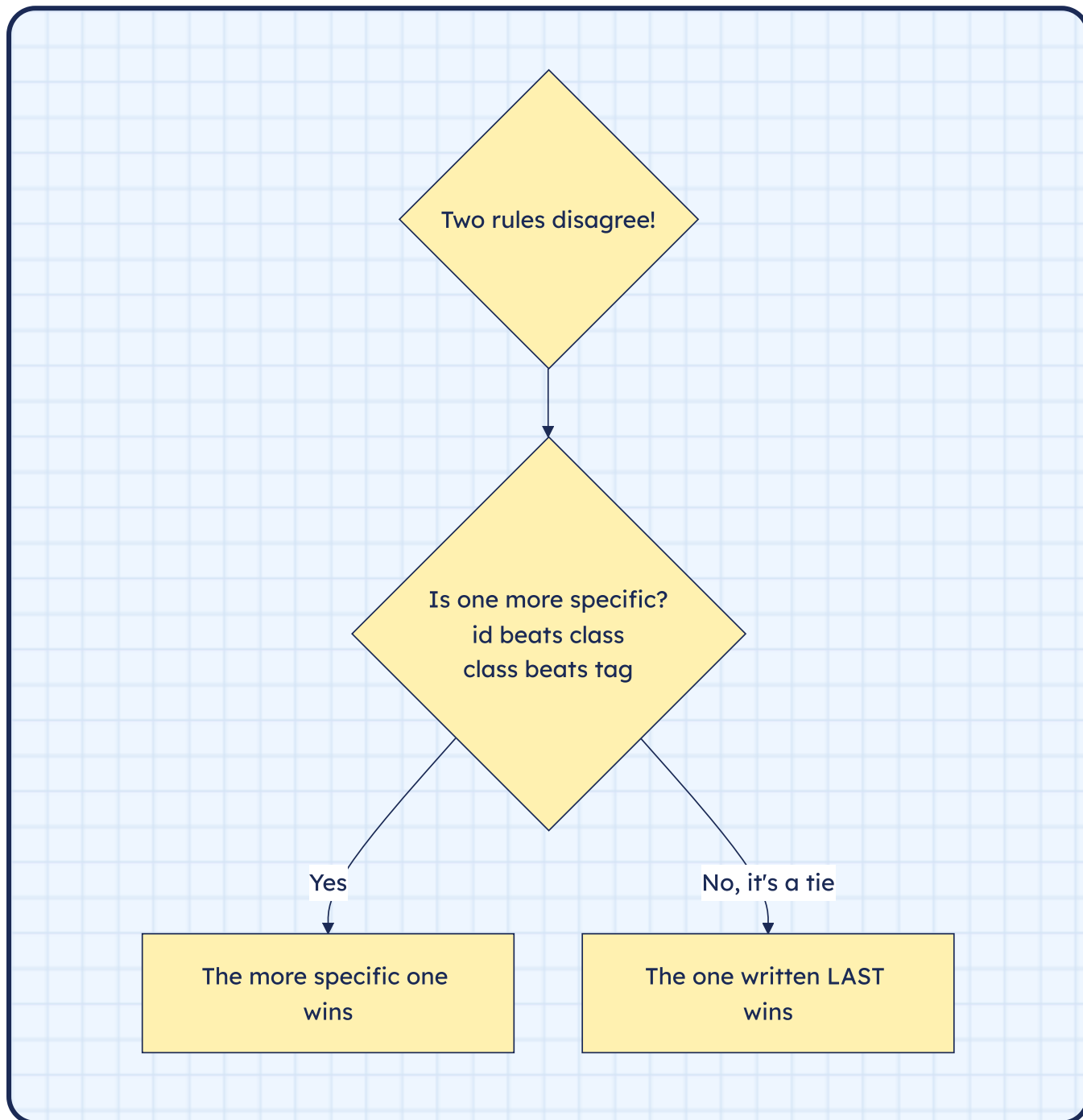


```
@keyframes hop {  
  0%   { translate: 0 0; }  
  50%  { translate: 0 -60px; }  
  100% { translate: 0 0; }  
}  
  
.pikachu {  
  animation: hop 1s ease-in-out infinite;  
}
```



When two rules argue

Sometimes two CSS rules want to change the same thing. One says red, one says blue. Who wins? The browser follows these steps, in this order.



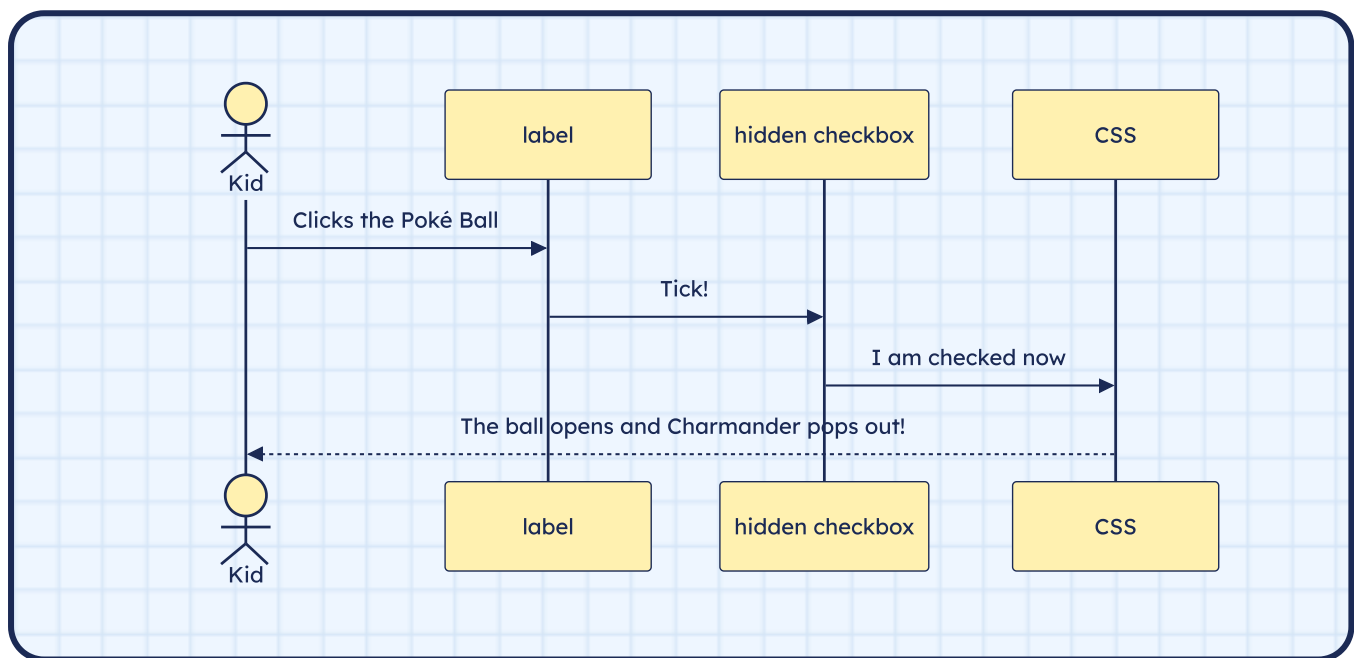
Try it: set up battles in the Specificity Showdown. Can ten classes beat one id?

Chapter 14

How AshTech's games remember



Our games use a clever trick. A `<label>` clicks a hidden checkbox for you. CSS notices the box is ticked with `:checked`, and changes the page. No JavaScript needed!

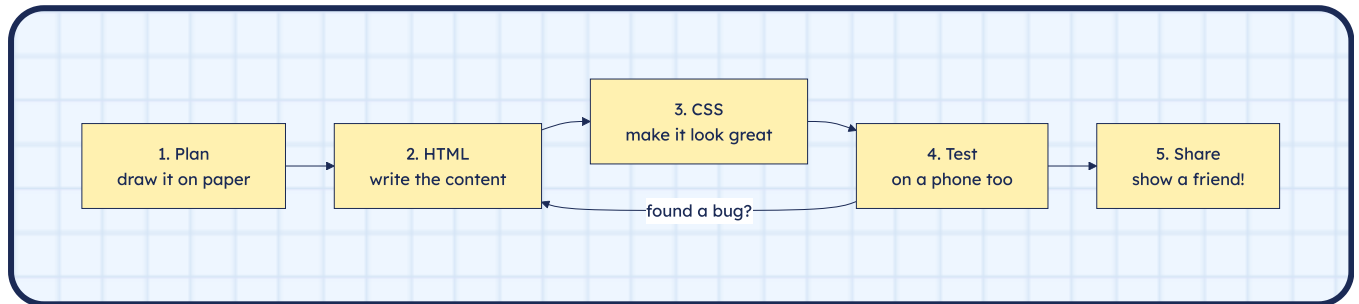


```
#p-charmander:checked + label .ball-top {
  rotate: -110deg;
}
```



Build your own page

Now it's your turn! Real web builders follow these steps. Don't worry about bugs: every builder finds them, and fixing them is part of the fun.

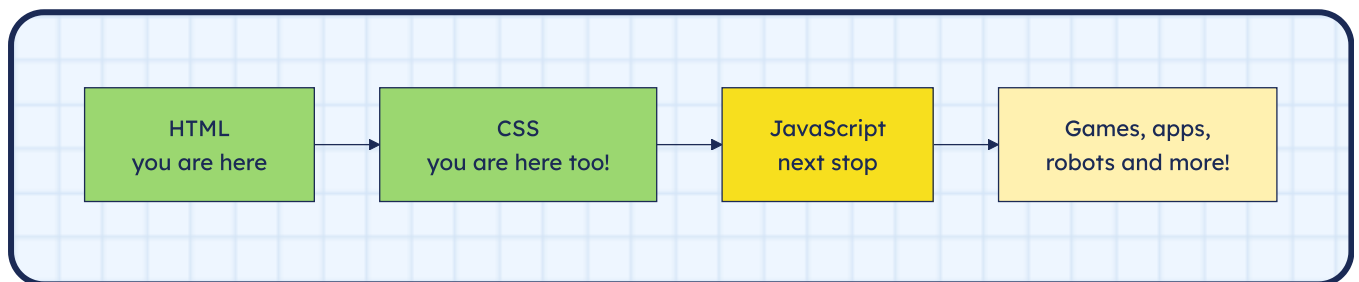


- ☐ My page has a `<title>`
- ☐ My page has one `<h1>`
- ☐ Every picture has `alt` words
- ☐ Every form box has a `<label>`
- ☐ I used at least one class in my CSS
- ☐ My page looks good on a phone



What comes next?

Once you love HTML and CSS, the next stop on the route is **JavaScript**. It's the language behind the Trainer Card download and the Workshop labs. After that? Apps, games, robots, even space rockets use code. Every expert was once a beginner, just like you.





Quick quiz

Answer these with a friend. Click a question to check your answer.

1. ► Which language says what things ARE?

HTML

2. ► What does the `/` in `</p>` mean?

It's a closing tag: the paragraph ends here.

3. ► Which list should you use for the steps of a recipe?

`<ol>`, because the order matters.

4. ► Why do pictures need `alt` words?

So people who can't see the picture can still know what it shows.

5. ► In `h1 { color: red; }`, which part is the selector?

`h1`

6. ► Which wins: `#partner` or `.fire`?

`#partner`. An id beats a class.

7. ► Name the four layers of the box model, inside to outside.

Content, padding, border, margin.

8. ► You want Pokémon cards in rows and columns. Flex or grid?

Grid!

Certificate of Web Building

This is to say that

.....

has finished the AshTech Kids' Handbook and can build web pages
with HTML and CSS!

Teacher's signature

Date